

SIGN LANGUAGE DETECTION AND VOICE ASSISTANCE SYSTEM

Course: Machine Learning

Submitted on: 30/11/2025

Presented by:

Hamza Ali Ahmed

Maruthi V Kamath

Manjushree M R

Prajwal

INTRODUCTION:

- Communication is difficult for people who cannot speak or see.
- Mute individuals use hand signs, but most people do not understand them.
- Visually impaired people cannot see hand signs at all.
- Our system uses a camera + machine learning model to recognize hand signs.
- The output is shown as text and also spoken using Text-to-Speech (TTS)

Problem statement:

People who cannot speak use hand signs to communicate, but most people do not understand sign language. Blind people also cannot see these signs. Because of this, it becomes difficult for mute and blind individuals to communicate with others. So, we need a system that can look at a hand sign, understand it, and convert it into text and speech to make communication easier for everyone.

GOAL

The main goal of this project is to create a system that can understand hand signs shown to a camera and convert them into text and speech so that people with speaking or seeing disabilities can communicate easily.

Objectives

- To detect hand signs in real time using a camera.
- To convert the detected sign into English text so others can understand the message.
- To convert the text into speech using Text-to-Speech so visually impaired users can hear the output.
- To build a simple and user-friendly website that anyone can use.
- To reduce communication barriers for mute, deaf, and visually impaired individuals.
- To provide a low-cost system using open-source tools.
- To make communication faster, easier, and more inclusive using machine learning.

Expected Outcomes:

- A working system that recognizes ASL signs.
- Real-time prediction using webcam.
- Speech output for visually impaired users.
- A low-cost, user-friendly tool for accessible communication.

Data Collection:

- ASL image datasets collected from Kaggle.
- Contains raw RGB images of single-hand ASL signs.
- Includes multiple samples of each alphabet sign (A–Z).
- Images used directly without preprocessing.

Description of the Data:

- All images contain one hand showing an ASL gesture.
- Dataset includes variations in lighting, angles, and hand shapes.
- Mediapipe used to extract 21 hand landmarks from each image.
- Each landmark has x, y, z coordinates → 63 numerical features.
- All extracted features saved into a CSV dataset for model training.

Data Preprocessing

- Images were passed through Mediapipe, which detects and normalizes hand landmarks.
- Ensured only valid frames with clear hand detection were included.
- Removed noisy or incomplete landmark samples.
- Verified that landmark coordinates were consistent across all images.
- Prepared a structured dataset of high-quality landmark values for training.

Feature Engineering

- Mediapipe extracted 21 hand landmarks per image.
- Each landmark contains x, y, z coordinates.
- Total features per image → 63 numerical values.
- These features capture hand geometry, finger orientation, and gesture structure.
- All extracted features were stored in a CSV dataset used for training the ML model.

Model Training:

- Used 63 landmark features extracted from Mediapipe.
- Created a structured CSV dataset for training.
- Chose Support Vector Machine (SVM) as the classification model.
- Trained using the RBF kernel for handling complex gesture patterns.
- Tuned key hyperparameters (C and Gamma) for best accuracy.
- Enabled probability=True for confidence scores.
- Saved the final trained model as svm_mediapipe.pkl.
- Model used for real-time webcam detection.

MODELS TESTED (MODEL SELECTION)

1. KNN

- Simple algorithm based on nearest neighbors.
- Slow prediction time for large datasets.
- Low accuracy due to 63-dimensional feature space.

2. Random Forest

- Better accuracy compared to KNN.
- Showed signs of overfitting on training data.
- Inconsistent predictions for visually similar gestures.

3. SVM (Selected Model)

- Best performance among all tested models.
- Excellent with numerical, landmark-based datasets.
- Strong at separating gestures with small differences.
- Fast and stable for real-time webcam detection.

JUSTIFICATION FOR SVM SELECTION

- Delivered the highest accuracy in our experiments.
- Performed consistently across all ASL hand signs.
- RBF kernel efficiently handled non-linear hand shapes.
- Did not overfit, unlike Random Forest.
- More stable and faster than KNN for real-time prediction.
- Ideal for structured feature vectors (63 points).
- Maintained accuracy even when gestures had minor variations.

EVALUATION & METRICS

Model Evaluation

- Tested on unseen ASL images to check generalization.
- Evaluated with accuracy, confusion matrix, and class-wise performance.

Key Observations

- High overall testing accuracy.
- Strong stability in real-time webcam inference.
- Misclassifications mainly occurred between M and N (very similar landmarks).

Performance Metrics Used

- Accuracy Score
- Confusion Matrix
- Real-Time Responsiveness

Results:

- SVM achieved high accuracy in ASL recognition.
- Real-time webcam detection worked smoothly.
- System produced correct text & speech output
- Mediapipe provided consistent and clean landmark features.
- System worked reliably under good lighting conditions.
- Predicted signs were shown as text on the interface.
- Text-to-Speech (TTS) produced clear and correct audio output.

Future Enhancements

- Add support for two-hand ASL gestures.
- Improve robustness: handle low light, cluttered background, and varied camera angles.
- Add continuous sign segmentation to separate words in fast signing.
- Improve real-time speed & stability.
- Build a mobile app (Android/iOS) for easier daily use.
- Provide multilingual TTS (Kannada, Hindi, English, etc.).
- Train on a larger, more diverse dataset (different skin tones, hand sizes, lighting).
- Edge deployment & optimization for small devices (quantization, pruning, faster inference).

Conclusion:

- Successfully built a real-time webcam-based ASL sign detection system.
- Used Mediapipe landmarks and SVM for accurate and stable recognition.
- System converts detected signs into text and speech for accessibility.
- Helps visually impaired users by providing clear audio output.
- Works smoothly in real time and is simple, low-cost, and user-friendly.
- Provides a strong base for future development into a full ASL translation system.

THANK YOU